MIL-C-50759A(MU)
15 May 1972
SUPERSEDING
MIL-T-50759(MU)
20 January 1972

MILITARY SPECIFICATION

COLLIMATOR & MOUNT : 10559901

1. SCOPE

1.1 This specification covers the Collimator & Mount 10559901 which consists of a collimator and mount. This equipment is referenced throughout this specification as collimator.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-O-13830	Optical Components of Fire Control Instruments: General Specification Governing the Manufacture, Assembly and Inspection of
MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-I-45607	Inspection Equipment, Acquisition, Disposition and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-810	Environmental Test Methods

DRAWINGS

U.S. Army, Frankford Arsenal

F10559901	Collimator & Mount: 10559901

FSC 4931

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PACKAGING DATA SHEET

10559901 Packaging of Collimator & Mount (10559901)

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Fabrication.- The collimator shall be manufactured in accordance with Drawing F10559901 and drawings pertaining thereto.

3.2 General specifications.- The contractor shall be responsible for adherence to, and compliance with, the following requirements of MIL-F-13926:

(a) Order of precedence
(b) Dimensions and tolerances
(c) Inorganic protective surface finishes
(d) Part identification and marking
(e) Workmanship

3.3 Environmental.

3.3.1 Storage temperatures.- The collimator shall show no evidence of physical failure such as cement separation, glass breakage or casting failure and shall meet all requirements of this specification at standard ambient temperature, +60°F to +90°F after having been exposed and thermally stabilized at ambient temperatures of -80°F and +160°F.

3.3.2 Operating temperatures.- The collimator shall meet the requirements of 3.7 and 3.8.1 while exposed to and thermally stabilized at ambient temperature of -40°F and +150°F.

3.3.3 Vibration.- The collimator inclosed in its carrying case, shall be vibrated while in a horizontal position at a constant frequency of 30 cycles per second with an amplitude of 1/16 inch (1/8 inch maximum excursion) for a period of 5 minutes plus or minus 15 seconds. Subsequent to vibration the collimator shall meet the requirements of 3.4 to 3.8.4 inclusive.

3.4 Sealing.- The collimator shall show no evidence of leakage in excess of 0.2 pounds per square inch gage when subjected to an internal pressure of 5 ± 0.10 pound per square inch (psig) for a minimum of 2 hours. The interior atmosphere of the accepted collimator shall contain nitrogen having a dew point at least as low as -25°F.

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3.5 Cleanliness.

3.5.1 Dirt.- Inspection for dirt and foreign particles (classed as dirt) shall be made from the objective end only. The maximum size of any visible particle, when the reticle is brought into sharp focus, shall be a reticle line width. The maximum number of dirt particles appearing within a central field of view of 50 mils diameter shall be 4. The maximum number of dirt particles appearing outside this field of view shall be 5. The minimum angular separation between dirt particles shall be 15 mils. Dirt particles smaller than 1/4 reticle line width shall be ignored regardless of distribution.

3.5.2 Other defects.- There shall be no evidence of moisture, grease, fingerprints, condensates, fractures, and adhesive separations when viewing thru the objective end of the instrument.

3.6 Accuracy.

3.6.1 Reticle accuracy.- The angular separation between the zero range line and the bottom of the 2500 meter, 0° (zero) cant block in elevation shall be 43.98 mils ± 0.05 mils.

3.6.2 Parallax.- The parallax between the reticle pattern at the 2000 meter point and the reticle of an observation telescope, which is parallax free for a target distance of 1000 ± 100 meters, shall not exceed 0.05 mils.

3.7 Lamp housing.- When exposed to the extreme temperature of 3.3.2, the lamp housing and lamp shall be removable and replaceable on the collimator.

3.8 Electrical.

3.8.1 Reticle illumination.- With the reticle light energized by a 24 volt power source, the reticle shall appear clearly defined when observed in a dark room.

3.8.2 Circuitry.- The circuitry of the collimator assembly shall be in accordance with the applicable electrical schematic drawings.

3.8.3 Dielectric strength.- There shall be no puncture of insulation, no sparkover, no decrease in dielectric strength, when the insulated areas of the circuitry is subjected to an alternating current (ac) root mean square (rms) potential of 1000 volts plus twice the rated voltage at a frequency of 60 cycles per second (cps) for a period of not more than 30 seconds nor less than 5 seconds.

3.8.4 Insulation resistance.- Unless otherwise specified on applicable drawings, all insulation between ungrounded circuits and ground, and between isolated circuits, shall have a minimum resistance of 100 megohms with 500 volts direct current (dc) applied.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 General provisions.- The component and subassembly inspection requirement of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 collimators and mounts. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

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4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the applicable Quality Assurance Provisions of MIL-F-13926.

4.3.2.2 Collimator and mount.- Examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plans specified in Tables I, II and III herein. Examination and tests for packaging, packing and marking shall be in accordance with Section 5 herein. The tabulated classification of defects in Tables I, II and III shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE I - 100% INSPECTION

NO.	CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
101.	Vibration	3.3.3	4.6.3, 4.6.3.1
102.	Sealing	3.4	4.6.4, 4.6.4.1
103.	Cleanliness	3.5	4.6.5
104.	Reticle accuracy	3.6.1	4.6.6
105.	Parallax	3.6.2	4.6.7
106.	Lamp housing	3.7	4.6.8
107.	Reticle illumination	3.8.1	4.6.9
108.	Circuitry	3.8.2	4.6.10
109.	Dielectric strength	3.8.3	4.6.11
110.	Insulation resistance	3.8.4	4.6.12

NOTE: The tests in Table I shall be conducted at a temperature between +60 and +90°F.

4.3.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105.

4.4 Special sampling.

4.4.1 General.- Collimators shall be selected at random by a Government Representative as a special sample from each produced. The sample shall meet the requirements and tests in Table II.

TABLE II

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
301.	Fabrication	3.3	Applicable drawings - visual
302.	General specification	3.2	MIL-F-13926 - visual

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4.4.2 Environmental.- Three collimators shall be selected at random by a Government representative as a special sample from each 50 produced or from each month's production, whichever occurs first. The samples shall meet the requirements and tests in Table III and shall then meet the requirements and tests in Table I.

TABLE III

NO.	CHARACTERISTICS	REQUIREMENTS	TEST PROCEDURES
303.	Storage temperature (-80°F, +160°F)	3.3.1	4.6.1
304.	Operating temperature (-40°F and +150°F)	3.3.2	4.6.2

4.4.3 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, care and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data listed in the List of Inspection Documents when provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment comparable with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerances used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when

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non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.2.3 Inspection equipment design requirements.

4.5.2.3.1 Sealing.- Special pressure testing equipment to test sealing of the collimator shall conform to the internal test requirements of Sheet 1 of Drawing F8565556, Pressure Tester-Two Station.

4.5.2.3.2 Reticle accuracy.- The test equipment utilized in this test shall be an angle reading telescope such as a theodolite or equivalent; capable of measuring angles in azimuth and elevation with an accuracy of at least ± .025 mils through a range of at least 75 mils. This angle reading telescope shall be capable of objective adjustment to view a target at infinity focus. A holding fixture shall be fabricated to hold the collimator assembly which will allow the collimator to be autocollimated to the angle reading telescope.

4.5.2.3.3 Set-up and calibration.- Rigidity stability, ease of maintenance, and adaptability of set-up and calibration equipment with complete provisions are also prime requirements. All set-up and calibration equipment necessary for certification of the testing equipment accuracy shall be furnished and will become a part of the testing equipment.

4.6 Test methods and procedures.

4.6.1 Storage temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.1. The collimator shall be exposed to each temperature extremes specified in 3.3.1 for a minimum of 5 hours. After each temperature exposures the collimator shall be stabilized in an ambient temperature between +60° and +90° Fahrenheit (F) and subjected to a visual and tactile examination to determine conformance with the requirements of 3.3.1. Caution shall be exercised during this test to avoid subjecting the collimator to thermal shock.

4.6.2 Extreme operating temperature.- The testing equipment utilized in this test shall be in accordance with the "Test Facilities" requirements of MIL-F-13926 and the conditions of 3.3.2. The collimator shall be exposed to each temperature extremes specified in 3.3.2 for a period of 5 hours. At the end of each 5 hour period, and while still at the temperature extremes, the collimator shall be subjected to the applicable tests of 4.6.8 and 4.6.9. The collimator shall conform to the requirements of 3.3.2. Caution shall be exercised during this test to avoid subjecting the collimator to thermal shock.

4.6.3 Vibration.- Vibrating equipment capable of providing the amplitude and frequency specified in 3.3.3 shall be utilized for this test. The vibrating equipment shall also be capable of vibrating the collimator along 3 mutually perpendicular axes while the collimator is

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maintained in a horizontal position. The collimator shall be horizontally mounted on the vibration table by means of an adapter. The collimator shall be vibrated in accordance with Procedure IX, Part 1 of MIL-STD-810, except that the frequency, frequency cycling, and amplitude of the vibration shall be as specified in 3.3.3. Upon completion of the vibration cycling the collimator shall be subjected to the tests specified in 4.6.3.1.

4.6.3.1 Examination.- Subsequent to vibration, the collimator shall be thoroughly examined for physical defects resulting from poor workmanship and materials to determine conformance with 3.3.3, prior to being subjected to the remaining tests specified herein. A visual examination shall be made for missing or damaged parts. A tactile examination shall also be conducted to determine looseness or improper functioning of moving parts. An audible examination shall be made of the collimator by shaking it and listening for loose internal parts or broken optical components, such as lenses and reticle, etc. Any one of the possible defects defined above shall be cause to consider a collimator defective. Collimator which have passed the above examinations shall be subjected to the test specified in 4.6.4 through 4.6.12 inclusive, to determine overall compliance with 3.3.3.

4.6.4 Sealing.- The internal pressurizing of the collimator to test sealing shall be performed with special testing equipment conforming to 4.5.2.3.1. The pressure shall be applied through the valve assembly. Pressurize the collimator with dry nitrogen gas to 5.0 ± 0.10 psig in accordance with the procedure for internal pressure testing of Sheet 1 of Drawing F8565556. The dry nitrogen gas shall have a dew point at least as low as minus 25°F. The applied pressure shall be retained for a minimum of 2 hours. Any evidence of leakage in the collimator at the end of the 2 hour time period, shall not exceed the tolerance specified in 3.4, as measured on the pressure gauge of the testing equipment.

4.6.4.1 Interior atmosphere.- Upon satisfactory completion of the sealing test the collimator shall be flushed with the dry nitrogen gas. The flushing shall be accomplished by opening the purging vent on the objective tube and allowing the nitrogen gas to flow through the collimator, until gas bled from the collimator conforms to the dew point requirements of 3.4, as measured with a dew point tester. Close the purging vent of the collimator.

4.6.5 Cleanliness.- Inspection for dirt or foreign particles shall be made by viewing through the objective end of the collimator by the viewing telescope to determine compliance with 3.5.1 and 3.5.2. Inspection for moisture shall be made by the technique of shadowing for detection of each optical surface.

4.6.6 Reticle accuracy.- This test shall be made utilizing the inspection equipment specified in 4.5.2.3.2 to determine compliance with 3.6.1. Place the collimator assembly in the test fixture specified in

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4.5.2.3.2 and adjust the azimuth and elevation adjustment to the center of their adjustment range. Adjust the viewing telescope objective for infinity focus. Adjust the diopter setting of the eyepiece, while viewing the viewing telescope reticle until reticle is in sharp focus. Set the angle reading viewing telescope to zero for elevation and azimuth. Adjust the test collimator assembly in azimuth and elevation until the reticles are super imposed. Adjust the viewing telescope to measure the angles specified in 3.6.1.

4.6.7 Parallax.- This test shall be performed utilizing the set-up in 4.6.6 with infinity focus in the viewing telescope. Parallax shall be recognized as any apparent displacement of the two reticles when the observer's head is moved from side to side, or up and down. Any movement between the reticle shall not exceed the requirement specified in 3.6.2.

4.6.8 Lamp housing.- The replacement of the lamp and lamp housing shall be accomplished by means of a tactile examination at the specified temperatures in accordance with the requirement of 3.7.

4.6.9 Reticle illumination.- The lamp assembly shall be energized by 24.0 volts to illuminate the collimator reticle. The illuminated etchings of each reticle pattern, when positioned in the focal plane, shall be clearly defined. Should the brightness of a reticle defect be greater than the brightness of a reticle line, the defect shall be cause for rejection.

4.6.10 Circuitry.- A continuity check of the collimator assembly shall be made utilizing standard testing equipment to determine conformance with the requirements specified in 3.8.2.

4.6.11 Dielectric strength.- This test shall be performed in accordance with MIL-STD-202, Method 301. Standard measuring equipment shall be used for all measurements. There shall be no evidence of failure or decrease in dielectric strength when subjected to the specified voltage for the time period indicated in 3.8.3.

4.6.12 Insulation resistance.- Insulation resistance shall be measured using Method 302, Test Condition B of MIL-STD-202 by means of standard measuring equipment. Resistance shall be measured at areas specified and shall meet the minimum resistance specified in conformance with the requirement of 3.8.4.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking.- Packaging, packing and marking shall be in accordance with Packaging Data Sheet 10559901. The level of protection shall be as specified in the procurement document.

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6. NOTES

6.1 Intended use.- The collimator is intended to provide a target for field check-out of Tank Fire Control Systems.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification
(b) Selection of applicable levels of preservation, packaging and packing (see 5.1)
(c) Applicable stock number

Custodian:

Army - MU

Preparing activity:

Army - MU

Project Number 4931-A104

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SPECIFICATION ANALYSIS SHEET

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INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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